

# HT Current Transformer Design

Design . No . : GCT0010265  
Date . : 7/21/2011

## WOUND PRIMARY TYPE

### SPECIFICATION :

H.S.V : 12 kV IL : 12/28/75 kVp STC : 25 kA For 1 sec. FREQ : 50 Hz.  
Ratio Type VA Class ISF<= ALF Vk>=(V) RCT(75°C)<=(ohm) Ie <=(mA)  
Core 1: 300-100/1 M 7.5 1.0  
Core 2: 300-100/1 P 7.5 5P 20

### MAGNETIC CORE DETAILS :

| Window (mm)     | Overall (mm) | BuiltUp (mm) | LM (cm) | Ht. (mm) | Aeff. (cm <sup>2</sup> ) | Wt. (kg) | Insulation : Amp.Turns | I/DSG/047 (R1) Core Material |
|-----------------|--------------|--------------|---------|----------|--------------------------|----------|------------------------|------------------------------|
| Core 1: 90 X 80 | 112 X 102    | 11.0         | 38.4    | 20.0     | 2.09                     | 0.61     | 400-1200               | CRGO-B                       |
| Core 2: 90 X 80 | 115 X 105    | 12.5         | 39.0    | 120.0    | 14.25                    | 4.25     | 400-1200               | CRGO-B                       |

### SECONDARY WINDING DETAILS :

|                              | Core 1   | Core 1  | Core 2  | Core 2  |
|------------------------------|----------|---------|---------|---------|
| Ratio                        | :100/1 A | 300/1 A | 100/1 A | 300/1 A |
| Nominal Current              | :1 A     | 1 A     | 1 A     | 1 A     |
| Secondary Cond               | :20 S    | 20 S    | 20 S    | 20 S    |
| Sec. C/S. (mm <sup>2</sup> ) | :0.6567  | 0.6567  | 0.6567  | 0.6567  |
| Cond Dia. (mm)               | :1.006   | 1.006   | 1.006   | 1.006   |
| No of Turns                  | :400     | 1200    | 400     | 1200    |
| Sec. Layers                  | :1.5     | 4.5     | 1.5     | 4.5     |
| Sec. Wind. Ht.(mm)           | :8.262   | 12.03   | 8.262   | 12.03   |
| Copper Wt.(Kg)               | :0.77    | 0.77    | 2.34    | 2.34    |
| RCT(75°C) (Ohms)             | :1.153   | 3.926   | 3.815   | 11.91   |
| I <sup>2</sup> R loss(W)     | :1.15    | 3.93    | 3.82    | 11.91   |
| Flux Density (Gauss)         | :4658    | 2050    | 893     | 511     |
| ISF                          | :4.51    | 10.24   |         |         |
| C1                           | :        |         |         |         |
| C2                           | :        |         |         |         |
| VK                           | :27      | 80      | 182     | 547     |
| Ie at I norm (A)             | :        |         | 0.003   | 0.001   |
| Ie at ALF (A)                | :        |         | 0.292   | 0.007   |
| % comp error(In)             | :        |         | 0.34    | 0.08    |
| % comp error(ALF)            | :        |         | 1.46    | 0.035   |
| Ie.(mA)                      | :        |         |         |         |
| SC.Den.(A/mm <sup>2</sup> )  | :380.7   | 126.9   | 380.7   | 126.9   |
| Sat.Den.(A/mm <sup>2</sup> ) | :58.96   | 51.95   | 121.49  | 116.75  |
| SC.Temp Rise(°C)             | :22.04   | 17.11   | 93.58   | 86.42   |

### PRIMARY WINDING DETAILS :

Cond.Style: Paper Covered -Double-Equal Wdg. Ht.  
Conductor : 10X0.8X(6)X2/10X0.8X(5)X2  
Cond/Coil : 10X0.8X(6)X1/10X0.8X(5)X1  
Primary Turns : 4  
Cross Section(mm<sup>2</sup>) : 176  
Wind.Ht.Side(mm) : 26  
Wind.Ht.Top(mm) : 26  
Wind.Width(mm) : 48  
I-Dynamic AT's : 250  
Nor.Curr.Den(A/mm<sup>2</sup>) : 1.7  
SCKt Curr.Den(A/mm<sup>2</sup>): 142.05 (For 1 sec)

### FORMER & DISTANCE PIECE :

Former Inside(mm) : 184 X 37  
Former Outside(mm) : 204 X 57  
Distance Piece(mm) : 15

### SECONDARY. WOUND. DIMENSIONS :

|         | Window (mm) | OverAll (mm) | HT (mm) | BUILDUP (mm) |
|---------|-------------|--------------|---------|--------------|
| CORE 1  | : 66 X 56   | 136 X 126    | 44      | 35           |
| CORE 2  | : 66 X 56   | 139 X 129    | 144     | 37           |
| MAX DIM | : 66 X 56   | 139 X 129    | 184     | 37           |

### CAST DIM. & CLEARANCES :

|            | Length  | Width    | Height | S.Len. |
|------------|---------|----------|--------|--------|
| Wind.(mm): | 256     | 139      | 191    | 184    |
| Mould No : | M-3539B | Type: T1 |        |        |
| Mould(mm): | 280     | 148      | 216    | 220    |
| Epoxy(mm): | 12      | 4        | 25     | 18     |

### WINDOW CLEARANCES :

Resin Width wise(mm): 9  
Resin Ht. Wise(mm) : 15

Drg No :PR-588(B)

### WEIGHTS :

Core (kg) : 4.86  
Sec.Cu.(kg): 3.11  
Pri.Cu.(kg): 4.83  
Epoxy(kg) : 14.00

### RATIO ERROR & PHASE ANGLE DISPLACEMENT CALCULATION:

Full :

| Ip     | Bm   | Re     | Ore    | Delta | %Ip  | bm   | Re     | Ore    | %Ip    | bm   | Re  | Ore    | Delta  | %Ip  | bm   | Re  | Ore    | Delta  |      |
|--------|------|--------|--------|-------|------|------|--------|--------|--------|------|-----|--------|--------|------|------|-----|--------|--------|------|
| 1.20   | 5590 | -0.981 | 0.019  | 2.99  | 1.20 | 2460 | -0.180 | 0.153  | 0.153  | 1.00 | 893 | -0.314 | -0.314 | 5.02 | 1.00 | 511 | -0.060 | -0.060 | 1.78 |
| 1.00   | 4658 | -1.063 | -0.063 | 3.53  | 1.00 | 2050 | -0.191 | 0.142  | 0.142  |      |     |        |        |      |      |     |        |        |      |
| 0.20   | 932  | -1.691 | -0.691 | 18.03 | 0.20 | 410  | -0.288 | 0.045  | 0.045  |      |     |        |        |      |      |     |        |        |      |
| 0.05   | 233  | -2.446 | -1.446 | 46.94 | 0.05 | 102  | -0.404 | -0.071 | -0.071 |      |     |        |        |      |      |     |        |        |      |
| Quarte |      |        |        |       |      |      |        |        |        |      |     |        |        |      |      |     |        |        |      |
| 1.20   | 1956 | -0.352 | 0.648  | 11.56 | 1.20 | 1249 | -0.083 | 0.250  | 0.250  |      |     |        |        |      |      |     |        |        |      |
| 1.00   | 1630 | -0.374 | 0.626  | 12.88 | 1.00 | 1041 | -0.086 | 0.247  | 0.247  |      |     |        |        |      |      |     |        |        |      |
| 0.20   | 326  | -0.466 | 0.534  | 25.59 | 0.20 | 208  | -0.101 | 0.232  | 0.232  |      |     |        |        |      |      |     |        |        |      |
| 0.05   | 81   | -0.499 | 0.501  | 46.23 | 0.05 | 52   | -0.102 | 0.231  | 0.231  |      |     |        |        |      |      |     |        |        |      |